



APRIL, 2025 A

REGIONAL MARITIME UNIVERSITY

DEPARTMENT: MECHANICAL ENGINEERING

BME/BMT/BEE 28 LEVEL 100 (REGULAR)

BME 107: MATERIAL SCIENCE

END OF FIRST SEMESTER EXAMINATION

INSTRUCTIONS : ANSWER ALL QUESTION IN SECTION A AND ANY TWO (2) QUESTIONS FROM SECTION B.

TIME : THREE (3) HOURS  
SECTION A (50 MARKS)

Answer ALL questions in this section (2 marks for each question)

*Write the letter that represents the correct answer in your answer booklet*

1. The earliest humans had access to the following materials that occur naturally EXCEPT
  - a. Biomaterials
  - b. Stone
  - c. Wood
  - d. Clay
2. Understanding Materials Science and Engineering should lead to the following EXCEPT
  - a. Proper materials usage
  - b. New design opportunities
  - c. Waste of materials
  - d. Right material for the right job
3. Sophisticated electronic devices rely on components that are made from?
  - a. Steels
  - b. Semiconducting materials
  - c. Cast irons
  - d. Plastics
4. When all the electrons occupy the lowest possible energies, an atom is said to be in its .....
  - a. Solid state
  - b. Ground state
  - c. Original state
  - d. Intermediate state
5. Which of the following is not a primary bond .....
  - a. Ionic bond
  - b. Covalent bond
  - c. Metallic bond
  - d. Hydrogen bond

|                    |                |              |                       |
|--------------------|----------------|--------------|-----------------------|
| Doc. No.: EXM/F/03 | Page No.: 1/5  | Issue No.: 1 | Last Update: 15/05/15 |
| Revision No.: 2    | Date: 12/04/06 | Author: EC   | Approved: PROVOST     |

6. For many alloy systems and at some specific temperature, there is a maximum concentration of solute atoms that may dissolve in the solvent to form a solid solution; this is called \_\_\_\_\_.
- Boundary line
  - Liquidus line
  - Solidus line
  - solubility limit
7. Line joining liquid phase with liquid and solid phase mixture is known as:
- Liquidus
  - Solidus
  - Tie line
  - Boundary
8. Line joining solid phase with liquid and solid phase mixture is known as:
- Liquidus
  - Solidus
  - Tie line
  - Boundary
9. Polymers are characterized by which of the following bonding type:
- Adhesive
  - Covalent
  - Ionic
  - Metallic
10. The modulus of elasticity indicates the \_\_\_\_\_ of a material.
- Hardness
  - Stiffness
  - Ductility
  - Brittleness
11. When \_\_\_\_\_ materials are used in machine members, impending failure is detected easily, and sudden failure is unlikely.
- Hard
  - Brittle
  - Stiff
  - Ductile
12. With an increase in temperature, hardness of material \_\_\_\_\_ and ductility \_\_\_\_\_.
- Increases, increases
  - Increases, decreases
  - Decreases, increases
  - Decreases, decreases
13. When the material is loaded within elastic limit, then the stress is \_\_\_\_\_ to strain.
- equal
  - directly proportional
  - inversely proportional
  - not equal

|                    |                |              |                       |
|--------------------|----------------|--------------|-----------------------|
| Doc. No.: EXM/F/03 | Page No.: 2/5  | Issue No.: 1 | Last Update: 15/05/15 |
| Revision No.: 2    | Date: 12/04/06 | Author: EC   | Approved: PROVOST     |

What is the coordination number for Face-centered cubic crystal structure?

a. 8  
b. 10  
c. 12  
d. 4

15. What is the coordination number for Body-centered cubic crystal structure?

a. 8  
b. 10  
c. 12  
d. 4

16. The main objective of annealing is .....

- a. To increase the hardness of the metal so that it can resist wear
- b. To enable it to cut other metals i.e. to make it suitable for cutting tools
- c. To soften the steel so that it may be easily machined or cold worked
- d. To reduce brittleness of the hardened steel and thus to increase ductility

17. The treatment in which the brittleness of martensite is reduced is called:

- a. Annealing
- b. Normalizing
- c. Quenching
- d. Tempering

*Write the most appropriate answer in your answer booklet*

Use The iron-iron carbide phase diagram (Figure 2 on page 6) to answer the following questions.

18. What is the melting temperature of pure iron (Fe)?

19. What is the melting temperature of Fe which contain 1 wt% C?

20. What is the melting temperature of Fe which contain 2 wt% C?

21. What is the melting temperature of Fe which contain 3 wt% C?

22. Write the Eutectic Reaction from the phase diagram.

23. What is the composition and temperature at which the eutectic reaction occurs?

24. Write the Eutectoid Reaction from the phase diagram.

25. What is the composition and temperature at which the eutectoid reaction occurs?

|                    |                |              |                       |
|--------------------|----------------|--------------|-----------------------|
| Doc. No.: EXM/F/03 | Page No.: 3/5  | Issue No.: 1 | Last Update: 15/05/15 |
| Revision No.: 2    | Date: 12/04/06 | Author: EC   | Approved: PROVOST     |

## SECTION B (50 MARKS)

Answer **ANY TWO (2)** questions from this section. **ALL** questions carry equal marks

### QUESTION ONE (25 MARKS)

- (a)
- i. Calculate the volume of an FCC unit cell in terms of the atomic radius  $R$ . (4 Marks)
  - ii. Show that the atomic packing factor for the FCC crystal structure is 0.74. (4 Marks)
  - iii. Copper has an atomic radius of 0.1278 nm, an FCC crystal structure, and an atomic weight of 63.5 g/mol. Given Avogadro's Number as  $6.023 \times 10^{23}$  atoms/mol compute its theoretical density. (4 Marks)
- (b) When the strength of a metal is cited for design purposes, the yield strength is used instead of the tensile strength. As an engineer do you agree with this statement? Give your explanation. (3 Marks)
- (c) Give an explanation as to why covalently bonded materials are generally less dense than ionically or metallically bonded ones. (3 Marks)
- (d) Describe the ionic bonding between magnesium and chlorine. (4 Marks)
- (e) Why are the eutectoid phase changes very important? (3 Marks)

### QUESTION TWO (25 MARKS)

- (a) Why do we study materials? (3 Marks)
- (b) Write the electron configuration for the following elements
- i. Nitrogen (N) (2 Marks)
  - ii. Silicon (Si) (2 Marks)
  - iii. Argon (Ar) (2 Marks)
- (c) A steel bolt in Fig. 1, 2.50 cm in diameter, carries a tensile load of 40 kN.

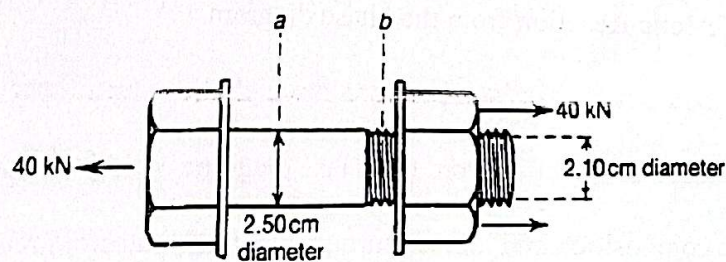


Figure 1

Estimate the average tensile stress at the (section a) and at the screwed (section b), where the diameter at the root of the thread is 2.10 cm. (10 Marks)

$$\text{stress} = \frac{F}{A}$$

|                    |                |              |                       |
|--------------------|----------------|--------------|-----------------------|
| Doc. No.: EXM/F/03 | Page No.: 4/5  | Issue No.: 1 | Last Update: 15/05/15 |
| Revision No.: 2    | Date: 12/04/06 | Author: EC   | Approved: PROVOST     |